

World's first search engine that finds connected 'things'

By indexing FTP and HTTP banners, Shodan lets users search for and find any Internet-connected object. This opens up opportunities to peep into refrigerators, theatres, museums, homes and even other computers

Having read an article about how somebody took a tour of Scottish castles using video footages found through Shodan (<http://www.shodan-hq.com/>), I too set about exploring this computer-finding search engine. I wanted to actually get a glimpse of what was happening in the Sydney Opera House, but had to satisfy myself with flashes of a museum in Japan since the Opera House cams were too well secured. I soon realised that viewing CCTV footages was just the tip of the iceberg. A lot more can be done with Shodan, such as finding the contents of other people's connected refrigerators, air-conditioners running at a particular temperature, computers running a software, mobile phones using a specific app, anonymous FTP servers, hosts affected by a new vulnerability and much more! More than that, you can find a lot of information about industrial control systems too.

Shodan is a hacker's tool. It is a search engine that lets you find specific objects connected to the Internet using a variety of filters. This means that unlike traditional Web search engines that help you find websites, Shodan helps you find computers, routers, cameras, phones, refrigerators, air-conditioners, ovens, traffic lights, sensors, locks and the increasing number of other things that are getting 'connected' of late.

Shodan's indexing is based mainly on banners, which is meta-data that the server sends back to the client. This can be information about the server software, options supported by the service, a welcome message or anything

else that the client would like to know before interacting with the server. For this reason, some also rightly call Shodan a public port scan directory or a search engine of banners.

While you can find a lot of things with Shodan, your search also has to be a little smarter, using the right search inputs, service filters and options. With proper filters, you can find devices based on city, country, latitude/longitude, hostname, operating system and IP. The results show the IP and physical location of each object found, the services running on the object, and more information. If the object is not properly secured, and you are smart enough, you can access its content too!

Exploits (<http://www.shodanhq.com/exploits/>), a variant of Shodan, lets you search for known vulnerabilities and exploits. You can search across Exploit DB, Metasploit, CVE, OSVDB and Packetstorm with one simple interface.

Well, now you know why Shodan is a tool for hackers. It is also a brilliant aid for penetration testers, to find and patch new vulnerabilities, and improve the security of industrial systems, sensor networks, etc. Shodan also provides official Shodan API documentation on how to access the search engine from Python, Ruby or Perl, to add its functionality to your programs too.

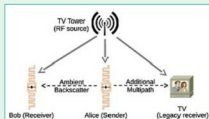
Call it the beginning of the Internet of Things if you will, but it is scary. For people who are already afraid of the privacy issues arising out of the IoT, Shodan is like supersonic fuel to their bad dreams. ●

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Read more about the Internet of Things at internetofthings.electronicshq.com

More use for thin air

University of Washington researchers recently presented a technology called Ambient Backscatter wherein TV, mobile and other radio frequency signals lingering in thin air can be used as both a source of power and a means of communication. Since this technology



does not require batteries, unlike other low-power near-field communication technologies, it is believed to be ideal for devices of the IoT era. For more information on this technology, you may read the paper at <http://www.theengineer.co.uk/Journals/2013/08/14/u/g/g/comm153-liu.pdf>

Embedded licenses

Embedded licensing is a new concept that ensures maximum returns from the smart software developed for IoT devices. An embedded license monitors two aspects of how the software is used: by controlling access to the software source code and by controlling how the software-embedded hardware is used. It protects software from IP theft, tampering and reverse-engineering. Learn more about embedded licenses from <http://www.safenet-inc.com/software-monetization/embedded-applications>

Wondering what's hot?



If you are looking for signs to decide what's hot and what's not, so you can base your own project on that, check out GitHub's new 'trending' page at <https://github.com/trending>. The page showcases the top-25 trending projects. The list is validated with a variety of data points including stars, forks, commits, follows, and page views, weighting them appropriately.